Problem · Unit 1 · 10 points · 60 min · Standard exam level

P14 — A spiral trajectory flown at constant speed: polar and intrinsic kinematics, a non-central force and zero power

*A drone P of mass m flies in a **horizontal** plane (its weight is balanced by the vertical lift, which is not studied). In this plane take the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ and the polar coordinates (r, φ) of P about O, with φ measured from $\mathbf{i}$ counter-clockwise and the polar basis $\mathbf{e}_r = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}$. The autopilot keeps P on the logarithmic spiral*

$$r = r_0\, e^{b\varphi} \qquad (r_0 > 0,\ b > 0)$$

*with a **constant speed** $|\mathbf{v}| = v_0$ and φ increasing. At $t = 0$ the drone is at $\varphi = 0$, $r = r_0$. The resultant of the horizontal forces on the drone (thrust and drag) is called $\mathbf{F}$. Let $\mathbf{e}_t$, $\mathbf{e}_n$ be the unit tangent and the principal normal of the trajectory, ρ its radius of curvature and α the angle between $\mathbf{e}_r$ and $\mathbf{e}_t$.*

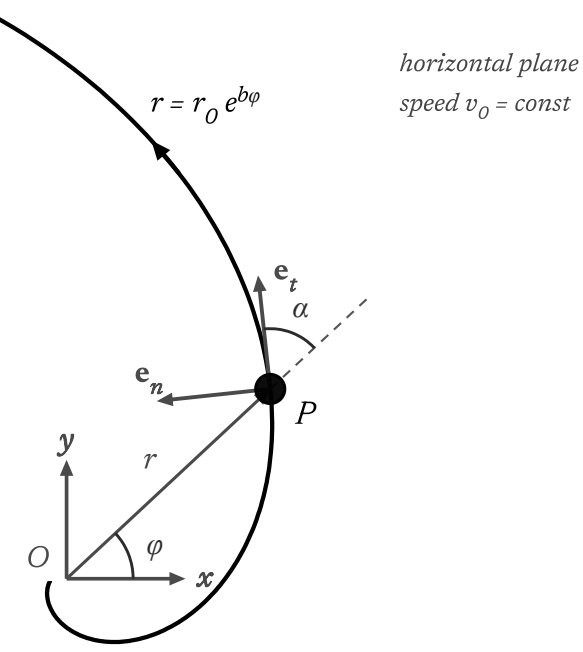


Figure 1. Drone P on the spiral $r = r_0 e^{b\varphi}$ flown with constant speed v_0 (φ increasing): polar coordinates (r, φ) about O, unit tangent $\mathbf{e}_t$ and principal normal $\mathbf{e}_n$, constant angle α between $\mathbf{e}_r$ and $\mathbf{e}_t$ (drawn for $b = 3/4$).

Setup

- Data: r_0, b, v_0, m. In the numerical parts: $r_0 = 20$ m, $b = 3/4$, $v_0 = 10$ m/s, $m = 2$ kg, and the drone flies from $r_1 = r_0 = 20$ m to $r_2 = 50$ m.
- Hypotheses: point particle, planar motion in the horizontal plane, S_0 inertial; the trajectory and the speed are imposed by the autopilot (the force $\mathbf{F}$ is whatever is needed to keep them).
- Useful constant: $k \equiv \sqrt{1+b^2}$; for $b = 3/4$, $k = \sqrt{1+9/16} = 5/4$.

Part (a) · 2 points

Velocity in polar coordinates and the constant angle

How many parameters fix the position of P on the trajectory? Write $\mathbf{v}$ in the polar basis and obtain $\dot r$ and $\dot\varphi$ as functions of r from the two conditions of the statement. Show that the angle α between the velocity and the radius is constant. Give the numbers for the data of the Setup.

Step 1. Parameters

A free point of the plane has two coordinates (r, φ). The trajectory $r = r_0 e^{b\varphi}$ is one relation between them, so one parameter (for instance φ, or r, or the arc length s) fixes the position on the curve. The speed law $|\mathbf{v}| = v_0$ then fixes how that parameter changes with time: nothing is left free, which is why this is a pure kinematics problem and $\mathbf{F}$ is an output.

Step 2. Velocity in the polar basis

The polar basis turns with φ; its relation to the inertial basis is

$$\begin{pmatrix}\mathbf{e}_r & \mathbf{e}_\varphi\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i} & \mathbf{j}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\cos^2\varphi+\sin^2\varphi & -\cos\varphi\sin\varphi+\sin\varphi\cos\varphi \\ -\sin\varphi\cos\varphi+\cos\varphi\sin\varphi & \sin^2\varphi+\cos^2\varphi\end{pmatrix} = I$ and $\det[R] = \cos^2\varphi+\sin^2\varphi = +1$ (same orientation as $\{\mathbf{i},\mathbf{j}\}$). Differentiating the rows with the chain rule gives $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$.

With $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$ (Notes §2.6), differentiating $\mathbf{r} = r\,\mathbf{e}_r$ with the product rule:

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\,\dot{\mathbf{e}}_r = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi$$

Step 3. The trajectory relates $\dot r$ and $\dot\varphi$

By the chain rule $\frac{d}{dt}r(\varphi(t)) = \frac{dr}{d\varphi}\dot\varphi$, with $\frac{dr}{d\varphi} = b\,r_0 e^{b\varphi} = b\,r$:

$$\dot r = b\,r\,\dot\varphi$$

Step 4. The speed fixes $\dot\varphi$

$$|\mathbf{v}|^2 = \dot r^2 + r^2\dot\varphi^2 = b^2r^2\dot\varphi^2 + r^2\dot\varphi^2 = (1+b^2)\,r^2\dot\varphi^2 = v_0^2$$

With $\dot\varphi > 0$ (the statement):

$$\dot\varphi = \frac{v_0}{r\sqrt{1+b^2}}, \qquad \dot r = b\,r\dot\varphi = \frac{b\,v_0}{\sqrt{1+b^2}}$$

The radial velocity is **constant** although $\dot\varphi$ is not: $\dot\varphi$ decreases as $1/r$.

Step 5. The angle with the radius

$\mathbf{v} = v_0\,\mathbf{e}_t$, so $\cos\alpha = \mathbf{e}_r\cdot\mathbf{e}_t = \dot r/v_0$ and $\sin\alpha = \mathbf{e}_\varphi\cdot\mathbf{e}_t = r\dot\varphi/v_0$:

$$\tan\alpha = \frac{r\dot\varphi}{\dot r} = \frac{r\dot\varphi}{b\,r\dot\varphi} = \frac{1}{b}$$

independent of the point: the logarithmic spiral cuts every radius at the same angle (it is the *equiangular* spiral).

Step 6. Numbers

$k = 5/4$: $\dot r = \frac{(3/4)\cdot 10}{5/4} = 6$ m/s, $r\dot\varphi = \frac{v_0}{k} = 8$ m/s (check: $6^2+8^2 = 10^2$), $\tan\alpha = 4/3$, $\alpha = 53.13^\circ$.

KEY RESULTS

- One parameter (φ, or r, or s); with the speed imposed, the motion is fully determined.
- $\dot r = b\,r\dot\varphi$, $\dot\varphi = \dfrac{v_0}{r\sqrt{1+b^2}}$, $\dot r = \dfrac{b\,v_0}{\sqrt{1+b^2}}$ **constant**.
- $\tan\alpha = \dfrac{r\dot\varphi}{\dot r} = \dfrac{1}{b}$; numbers: $\dot r = 6$ m/s, $r\dot\varphi = 8$ m/s, $\alpha = \arctan(4/3) = 53.13^\circ$.

Part (b) · 1.5 points

Time law: $r(t)$, $\varphi(t)$ and the flight time

Integrate the results of (a) to obtain $r(t)$ and $\varphi(t)$, and check that they lie on the spiral. Compute the time needed to go from $r_1 = 20$ m to $r_2 = 50$ m and the angle turned.

Step 1. Radial law

$\dot r = c$ with $c = b v_0/k$ constant. Integrating from 0 to t with Barrow's rule, $r(t) - r(0) = \int_0^t c\,d\tau = c\,t$:

$$r(t) = r_0 + \frac{b\,v_0}{\sqrt{1+b^2}}\,t$$

Step 2. Angular law (quadrature)

$$\varphi(t) = \varphi(0) + \int_0^t \frac{v_0}{k\,(r_0 + c\tau)}\,d\tau$$

Substitution $u = r_0 + c\tau$, $du = c\,d\tau$; the limits $\tau = 0 \mapsto u = r_0$ and $\tau = t \mapsto u = r(t)$:

$$\varphi(t) = \frac{v_0}{k\,c}\int_{r_0}^{r(t)} \frac{du}{u} = \frac{v_0}{k\,c}\Big[\ln u\Big]_{r_0}^{r(t)} = \frac{v_0}{k\,c}\big(\ln r(t) - \ln r_0\big) = \frac{1}{b}\ln\frac{r(t)}{r_0}$$

since $\frac{v_0}{kc} = \frac{v_0}{k}\cdot\frac{k}{b v_0} = \frac{1}{b}$. Inverting, $r(t) = r_0 e^{b\varphi(t)}$: the time law lies on the spiral, as it must.

Step 3. Flight time and angle turned

$r_2 - r_1 = c\,t_{12}$:

$$t_{12} = \frac{(r_2 - r_1)\sqrt{1+b^2}}{b\,v_0} = \frac{30\cdot(5/4)}{(3/4)\cdot 10} = 5\ \text{s}$$

$$\Delta\varphi = \frac{1}{b}\ln\frac{r_2}{r_1} = \frac{4}{3}\ln 2.5 = 1.2217\ \text{rad} = 70.00^\circ$$

KEY RESULTS

- $r(t) = r_0 + \dfrac{b\,v_0}{\sqrt{1+b^2}}\,t,\ \varphi(t) = \dfrac{1}{b}\ln\dfrac{r(t)}{r_0}.$
- $t_{12} = \dfrac{(r_2 - r_1)\sqrt{1+b^2}}{b\,v_0} = 5\ \text{s};\ \Delta\varphi = \dfrac{\ln(5/2)}{3/4} = 1.2217\ \text{rad} = 70.00^\circ.$

Part (c) · 1.5 points

Arc length and the intrinsic basis

Obtain the arc length $s(r)$ measured from r_0, and check it against $v_0\,t_{12}$. Write $\mathbf{e}_t$ and the principal normal $\mathbf{e}_n$ in the polar basis.

Step 1. Arc length

From $\varphi(r) = \frac{1}{b}\ln\frac{r}{r_0}$, $\frac{d\varphi}{dr} = \frac{1}{br}$, so $r\,d\varphi = \frac{dr}{b}$ and

$$ds = \sqrt{dr^2 + r^2 d\varphi^2} = \sqrt{1 + \frac{1}{b^2}}\,dr = \frac{\sqrt{1+b^2}}{b}\,dr$$

Integrating from r_0 to r (Barrow):

$$s(r) = \int_{r_0}^{r} \frac{k}{b}\,dr' = \frac{k}{b}\Big[r'\Big]_{r_0}^{r} = \frac{\sqrt{1+b^2}}{b}\,(r - r_0)$$

From 20 m to 50 m: $s = \frac{5/4}{3/4}\cdot 30 = 50$ m, equal to $v_0 t_{12} = 10\cdot 5 = 50$ m, as it must at constant speed.

Step 2. Unit tangent

$\mathbf{e}_t = \mathbf{v}/v_0$ with the components of (a):

$$\mathbf{e}_t = \frac{\dot r}{v_0}\,\mathbf{e}_r + \frac{r\dot\varphi}{v_0}\,\mathbf{e}_\varphi = \frac{b\,\mathbf{e}_r + \mathbf{e}_\varphi}{\sqrt{1+b^2}} = \cos\alpha\,\mathbf{e}_r + \sin\alpha\,\mathbf{e}_\varphi$$

Step 3. Principal normal

The components of $\mathbf{e}_t$ are constant, so only the basis vectors change. With the product rule and $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$, $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$:

$$\dot{\mathbf{e}}_t = \frac{b\,\dot\varphi\,\mathbf{e}_\varphi - \dot\varphi\,\mathbf{e}_r}{k} = \dot\varphi\,\frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{k}$$

By the chain rule $\frac{d\mathbf{e}_t}{ds} = \frac{\dot{\mathbf{e}}_t}{\dot s}$ with $\dot s = v_0$, and the principal normal is the unit vector of $d\mathbf{e}_t/ds$ (Notes §2.4). The vector $(-\mathbf{e}_r + b\,\mathbf{e}_\varphi)/k$ has modulus $\sqrt{1+b^2}/k = 1$, so

$$\mathbf{e}_n = \frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}}, \qquad \left|\frac{d\mathbf{e}_t}{ds}\right| = \frac{\dot\varphi}{v_0}$$

Check: $\mathbf{e}_t \cdot \mathbf{e}_n = (-b+b)/k^2 = 0$, and $\mathbf{e}_n$ is $\mathbf{e}_t$ turned $+90^\circ$ (to the left of the motion, towards the concave side and partly towards O).

KEY RESULTS

- $s(r) = \dfrac{\sqrt{1+b^2}}{b}\,(r - r_0)$; from 20 m to 50 m: $s = 50\ \text{m} = v_0 t_{12}$.
- $\mathbf{e}_t = \dfrac{b\,\mathbf{e}_r + \mathbf{e}_\varphi}{\sqrt{1+b^2}}$, $\mathbf{e}_n = \dfrac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}}$ (towards the concave side, to the left of the motion).

Part (d) · 2 points

Acceleration, radius of curvature

Compute the acceleration in polar components. Obtain the radius of curvature ρ and show that $\mathbf{a} = \dfrac{v_0^2}{\rho}\,\mathbf{e}_n$. Numbers at $r = 20$ m and at $r = 50$ m.

Step 1. Polar acceleration

Differentiating $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi$ with the product rule (Notes §2.6):

$$\mathbf{a} = (\ddot r - r\dot\varphi^2)\,\mathbf{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\,\mathbf{e}_\varphi$$

Step 2. The derivatives needed

$\ddot r = 0$ because $\dot r$ is constant. From $\dot\varphi = \frac{v_0}{k}\, r^{-1}$, with the chain rule $\frac{d}{dt} r^{-1} = -r^{-2}\dot r$:

$$\ddot\varphi = -\frac{v_0}{k}\,\frac{\dot r}{r^2} = -\frac{\dot r\,\dot\varphi}{r}$$

Step 3. Components

$$a_r = 0 - r\dot\varphi^2 = -r\,\frac{v_0^2}{r^2 k^2} = -\frac{v_0^2}{r(1+b^2)}$$

$$a_\varphi = r\Big(-\frac{\dot r\dot\varphi}{r}\Big) + 2\dot r\dot\varphi = \dot r\dot\varphi = \frac{b v_0}{k}\cdot\frac{v_0}{rk} = \frac{b\,v_0^2}{r(1+b^2)}$$

Step 4. Intrinsic form and radius of curvature

The curvature is $1/\rho = |d\mathbf{e}_t/ds| = \dot\varphi/v_0 = \frac{1}{rk}$ (part (c)), so $\rho = r\sqrt{1+b^2}$. In the intrinsic basis (Notes §2.5) $\mathbf{a} = \dot v\,\mathbf{e}_t + \frac{v^2}{\rho}\,\mathbf{e}_n$ with $\dot v = 0$:

$$\mathbf{a} = \frac{v_0^2}{r\sqrt{1+b^2}}\,\mathbf{e}_n = \frac{v_0^2}{r\sqrt{1+b^2}}\cdot\frac{-\mathbf{e}_r + b\,\mathbf{e}_\varphi}{\sqrt{1+b^2}} = -\frac{v_0^2}{r(1+b^2)}\,\mathbf{e}_r + \frac{b\,v_0^2}{r(1+b^2)}\,\mathbf{e}_\varphi$$

which coincides with Step 3: the two routes agree. Check: $\sqrt{a_r^2 + a_\varphi^2} = \frac{v_0^2\sqrt{1+b^2}}{r(1+b^2)} = \frac{v_0^2}{\rho}$.

Step 5. Numbers

At $r = 20$ m: $\rho = 20\cdot\frac{5}{4} = 25$ m, $|\mathbf{a}| = \frac{100}{25} = 4$ m/s^2, $a_r = -\frac{100}{20\cdot 25/16} = -3.2$ m/s^2, $a_\varphi = \frac{3}{4}\cdot 3.2 = 2.4$ m/s^2 (check: $3.2^2 + 2.4^2 = 16$). At $r = 50$ m: $\rho = 62.5$ m and $|\mathbf{a}| = 1.6$ m/s^2: the spiral opens up and the turn becomes gentler.

KEY RESULTS

- $a_r = -\dfrac{v_0^2}{r(1+b^2)},\ a_\varphi = \dfrac{b\,v_0^2}{r(1+b^2)},\ \rho = r\sqrt{1+b^2},\ |\mathbf{a}| = \dfrac{v_0^2}{r\sqrt{1+b^2}}.$
- At $r = 20$ m: $\rho = 25$ m, $a_r = -3.2$ m/s^2, $a_\varphi = 2.4$ m/s^2, $|\mathbf{a}| = 4$ m/s^2; at $r = 50$ m: $|\mathbf{a}| = 1.6$ m/s^2.

Part (e) · 1.5 points

Force, torque and angular momentum about O

Find the horizontal force $\mathbf{F}$ needed at $r = 20$ m. Is it a central force about O? Compute the torque $\mathbf{M}_O$ and the angular momentum $\mathbf{H}_O(t)$, and verify the angular momentum theorem between r_1 and r_2.

Step 1. The force

Newton's second law in the inertial frame, $\mathbf{F} = m\mathbf{a}$, with $m = 2$ kg and the numbers of (d):

$$\mathbf{F} = 2\,(-3.2\,\mathbf{e}_r + 2.4\,\mathbf{e}_\varphi) = -6.4\,\mathbf{e}_r + 4.8\,\mathbf{e}_\varphi \ \text{N}, \qquad |\mathbf{F}| = 8 \ \text{N}$$

In general $\mathbf{F} = \frac{m v_0^2}{\rho}\mathbf{e}_n$. Since $\mathbf{e}_n \cdot (-\mathbf{e}_r) = 1/k$, the angle γ between $\mathbf{F}$ and $-\mathbf{e}_r$ has $\cos\gamma = 1/\sqrt{1+b^2}$, i.e. $\tan\gamma = b$: $\gamma = \arctan(3/4) = 36.87^\circ$. The force is **not** central: $F_\varphi \neq 0$ at every point.

Step 2. Torque

$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\,\mathbf{e}_r \times (F_r\mathbf{e}_r + F_\varphi\mathbf{e}_\varphi) = r\,F_\varphi\,\mathbf{k}$ (since $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$):

$$\mathbf{M}_O = m\,r\,a_\varphi\,\mathbf{k} = m\,r\,\frac{b\,v_0^2}{r(1+b^2)}\,\mathbf{k} = \frac{m\,b\,v_0^2}{1+b^2}\,\mathbf{k} = \frac{2\cdot 0.75\cdot 100}{1.5625}\,\mathbf{k} = 96\,\mathbf{k}\ \text{N m}$$

constant along the whole flight, although $\mathbf{F}$ changes.

Step 3. Angular momentum

$\mathbf{r}\times\mathbf{v} = r\,\mathbf{e}_r \times (\dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi) = r^2\dot\varphi\,\mathbf{k}$, so with $\dot\varphi$ of (a):

$$\mathbf{H}_O = m\,r^2\dot\varphi\,\mathbf{k} = \frac{m\,r(t)\,v_0}{\sqrt{1+b^2}}\,\mathbf{k}$$

At $r = 20$ m: $H_O = \frac{2\cdot 20\cdot 10}{1.25} = 320$ kg m^2/s; at $r = 50$ m: 800 kg m^2/s.

Step 4. Angular momentum theorem

About the fixed point O, $\dot{\mathbf{H}}_O = \mathbf{M}_O$ (Notes §4.8). By the chain rule $\frac{d}{dt} H_O = \frac{m v_0}{k}\dot r = \frac{m v_0}{k}\cdot\frac{b v_0}{k} = \frac{m\,b\,v_0^2}{1+b^2} = M_O$. Integrating from 0 to t_{12} (Barrow): $H_O(t_{12}) - H_O(0) = \int_0^{t_{12}} M_O\,dt = 96\cdot 5 = 480$ kg m^2/s, and indeed $800 - 320 = 480$. The angular momentum is **not** conserved, and the areal velocity $\frac{1}{2}r^2\dot\varphi$ grows: Kepler's second law does not hold for this force.

KEY RESULTS

- At $r = 20$ m: $F_r = -6.4$ N, $F_\varphi = 4.8$ N, $|\mathbf{F}| = 8$ N; it makes $\arctan b = 36.87^\circ$ with $-\mathbf{e}_r$: **not central**.
- $\mathbf{M}_O = m\,r\,a_\varphi\,\mathbf{k} = \dfrac{m\,b\,v_0^2}{1+b^2}\,\mathbf{k} = 96$ N m (constant); $\mathbf{H}_O = \dfrac{m\,r(t)\,v_0}{\sqrt{1+b^2}}\,\mathbf{k}$: 320 kg m^2/s at 20 m, 800 kg m^2/s at 50 m; $\Delta H_O = 96\cdot 5 = 480$ kg m^2/s.

Part (f) · 1.5 points

Power, kinetic energy and the trap

Compute the power of $\mathbf{F}$ and the work it does between r_1 and r_2, and check the result with the kinetic energy theorem. A student argues: "the speed is constant, so the drone is in equilibrium and the net force is zero; and since $\mathbf{F}$ does no work, it must be a central (or conservative) force". Discuss.

Step 1. Power and work

The power of $\mathbf{F}$ is $P = \mathbf{F}\cdot\mathbf{v}$ (Notes §4.5). With $\mathbf{F} = \frac{m v_0^2}{\rho}\mathbf{e}_n$ and $\mathbf{v} = v_0\mathbf{e}_t$:

$$P = \frac{m\,v_0^3}{\rho}\,(\mathbf{e}_n\cdot\mathbf{e}_t) = 0$$

In polar components the same result is $m(a_r\dot r + a_\varphi r\dot\varphi) = m\frac{v_0^2}{r k^2}\big(-\frac{b v_0}{k} + b\,\frac{v_0}{k}\big) = 0$. The work between r_1 and r_2 is $W_{12} = \int_0^{t_{12}} P\,dt = 0$.

Step 2. Kinetic energy theorem

$\frac{d}{dt}\big(\frac{1}{2}m\,\mathbf{v}\cdot\mathbf{v}\big) = m\,\mathbf{a}\cdot\mathbf{v} = P = 0$, consistent with $|\mathbf{v}| = v_0$: $E_{kin} = \frac{1}{2}\cdot 2\cdot 10^2 = 100$ J at every instant.

Step 3. The trap

- "Constant speed means zero force" confuses the speed $|\mathbf{v}|$ with the velocity vector. The direction of $\mathbf{v}$ turns with angular rate $\dot\varphi$ (since α is constant, $\mathbf{e}_t$ turns exactly as $\mathbf{e}_r$), so $\mathbf{a} = \frac{v_0^2}{\rho}\mathbf{e}_n \neq \mathbf{0}$: at 20 m the autopilot must supply 8 N.
- "No work, so central" is false: zero power only means $\mathbf{F}\perp\mathbf{v}$ at each instant. Here $\mathbf{F}$ is perpendicular to $\mathbf{v}$ but makes 36.87° with the radius, so it has a moment about O and $\mathbf{H}_O$ grows by 480 kg m^2/s.
- A force perpendicular to the velocity is not conservative in general either: being workless along **this** path says nothing about the work along other paths. What matters for the energy budget is that it does no work along the actual motion.

KEY RESULTS

- $P = \mathbf{F}\cdot\mathbf{v} = \frac{mv_0^2}{\rho}\, v_0\,(\mathbf{e}_n\cdot\mathbf{e}_t) = 0$; $W_{12} = 0$; $E_{kin} = \frac{1}{2}mv_0^2 = 100$ J at every instant.
- Constant **speed** is not constant **velocity**: $|\mathbf{F}| = mv_0^2/\rho \neq 0$. Zero power only says $\mathbf{F}\perp\mathbf{v}$; the force is neither zero nor central ($\mathbf{M}_O \neq \mathbf{0}$).

Interpretation and checks

- **Dimensional check:** $\dot r = bv_0/k$ and $r\dot\varphi$ are m/s; $\rho = rk$ is m; $a = v_0^2/\rho$ is $\mathrm{m^2 s^{-2}/m}$ = $\mathrm{m\ s^{-2}}$; $M_O = mbv_0^2/k^2$ is $\mathrm{kg\ m^2 s^{-2}}$ = N m; $H_O = mrv_0/k$ is $\mathrm{kg\ m^2/s}$; $t_{12} = (r_2 - r_1)k/(bv_0)$ is s.
- **Limiting cases:** $b \to 0$ gives $r = r_0$, $\dot r = 0$, $\alpha = 90^\circ$, $\rho = r_0$, $a_\varphi = 0$: uniform circular motion with a central force $-mv_0^2/r_0\,\mathbf{e}_r$ and $\mathbf{H}_O$ constant. $b \to \infty$ gives $\alpha \to 0$: an almost radial straight flight with $\rho \to \infty$ and $\mathbf{a} \to \mathbf{0}$.
- **Flying inwards:** with φ decreasing, r decreases linearly and reaches O after the finite time $r_0k/(bv_0) = 10/3$ s and the finite length $r_0k/b = 100/3$ m, while $\varphi = \frac{1}{b}\ln(r/r_0) \to -\infty$: infinitely many turns in a finite time, with $\dot\varphi$ and $|\mathbf{a}|$ growing without bound. A real autopilot cannot follow the spiral to the end.
- **Common mistakes in the exam:** (1) setting $a_r = 0$ because $\dot r$ is constant (the centripetal term $-r\dot\varphi^2$ remains); (2) taking $\dot\varphi$ constant and dropping $r\ddot\varphi$ in a_φ; (3) writing $\rho = r$ as for a circle; (4) concluding from $P = 0$ that $\mathbf{F} = \mathbf{0}$ or that $\mathbf{H}_O$ is conserved.